liquid-drop model and obtained the "double-well" potential energy curve in Figure 11.14b. In the first well the nucleus is a spheroid with the major axis about 25% larger than the minor. In the second well, the deformation is much larger, the axis ratio being about 1.8. A nucleus in the second well is metastable (i.e. in isomeric state) as it is unstable to γ-decay to the first well or to fission. Fission from the second well is hindered by a 2−3 MeV barrier, while from the first well the barrier is 5−6 MeV, accounting for the difference in half-lives.

The single-well curve in Figure 11.14a predicts symmetric fission whereas the double-well curve (Fig. 11.14b) leads to the correct prediction of asymmetric fission and a thin neck. Incorporation of shell effects in the fission model also leads to the prediction that the half-lives of very heavy nuclides ($Z \geq 106$) must be longer than the simple liquid-drop model would indicate. This led to a search for "super heavy" elements with $Z = 110 - 118$. The major difficulty is the release of reaction energy which causes immediate fission of the compound nucleus. Hence, strongly endothermic reactions have been tried. However, these have extremely small reaction yields. The heaviest man made elements are discussed in Ch. 14.

11.8. PHOTONUCLEAR REACTIONS

If a photon transfers sufficient energy to a nucleus to excite it to a higher state, three possibilities for de-excitation exist: (a) the same energy is immediately re-emitted isotropically, (b) a long-lived isomer may be formed which decays through emission of one or more γ-rays, and (c) the nucleus disintegrates. The first process (a) is referred to as the Mössbauer effect and is discussed in chapter 7. The second process (b) of nuclear de-excitation has also been discussed earlier. The third process (c) is referred to as *photonuclear disintegration*. The energy transferred to the nucleus must be sufficient to excite it above the dissociation energy for a proton, neutron, or other particle. A large energy transfer can also induce fission of heavy nuclei, *photo fission*.

The simplest photodisintegration process is that of the deuteron, whose binding energy is 2.23 MeV. If the γ-ray energy absorbed exceeds this value, a neutron and a proton are formed. This is a common reaction in nuclear reactors using heavy water as moderator, because the fission γ-ray energy is often several MeV. However, γ-rays of such energy rarely occur in the radioactive decay of nuclides (i.e. with half-lives of hours or longer). The cross-section for photodisintegration of ^{2}H has a maximum value of 2.4 mb at 4.3 MeV E_γ.

The energy necessary for photodisintegration of a nucleus is calculated from known nuclear masses. It is obviously easier to remove one particle than several from a nucleus. As a result we find that E_γ must be ≥ 5 MeV for photodisintegration of heavier nuclei.

For $10 \leq E_\gamma \leq 40$ MeV the photon wavelength is comparable to the nuclear size. It is therefore easily absorbed, which causes collective nuclear vibrational motions (so-called dipole vibrations, because the neutrons and protons are assumed to vibrate in separate groups). De-excitation occurs through γ-emission. This is known as the "giant resonance" region, because the total cross-section for heavier nuclides goes up to hundreds of millibarns. For higher E_γ, nucleons may be expelled, the main reactions being (γ,n), (γ,2n) and (γ,np) in order of descending importance. As the γ-energy increases and the wavelength decreases to nucleon dimensions, interaction with nuclear groups (e.g. deuterons) or single nucleons takes place. Below 550 MeV one pion plus a nucleon may be emitted in the de-excitation following the photon absorption. At higher energies, several pions may be emitted. Very little is known about the details of these processes.

11.9. EXERCISES

11.1. A 0.01 mm thick gold foil, 1 cm^2 in area, is irradiated with thermal neutrons. The (n,γ) cross-section is 99 b. What is the transformation rate at a n-flux of 10^{19} n m^{-2} s^{-1}?

11.2. Assume the irradiation time of the gold foil in the previous problem is one week. What percentage of the original gold atoms in the target have undergone transformation?

11.3. A water-cooled copper foil (0.1 mm thick) is irradiated by the internal beam of a sector focused cyclotron with 1.2 mA H^+ ions of 24 MeV for 90 min. The reaction ^{63}Cu(p,pn)^{62}Cu occurs with a probability of 0.086 b. Copper consists to 69% of ^{63}Cu. The proton beam has a cross-section of only 15 mm^2. (a) How many ^{62}Cu atoms have been formed? (b) What fraction of the projectiles have reacted to form ^{62}Cu? (c) What cooling effect is required (kW) at the target?

11.4. Calculate the macroscopic cross-section for reaction of natural uranium with thermal neutrons. See Figures 14.1 and 19.5.

11.5. (a) Estimate the yield (% of fissions) of ^{142}La in thermal fission of ^{235}U given a chain yield of 5.839% for A = 142, most probable charge = 55.86 and a width parameter of 0.56. (b) Is an appreciable amount of ^{142}Nd formed directly in thermal fission of ^{235}U?

11.6. The total kinetic energy of the fragments from thermal fission of ^{239}Pu is 177.7 MeV and the average fragment masses are 100.34 and 139.66 u respectively. What are the kinetic energies of the average light and heavy mass fragments?

11.7. Calculate the kinetic energy of the ^{4}He ion formed through thermal neutron capture in ^{10}B.

11.8. What is the minimum photon energy required for the reaction ^{11}B(γ,n)^{10}B?